IPDE-Extended Bivariate CRM

Thall-Cook and Braun Joint Binary Models

Technical specification for model development and simulation

Purpose. This document describes an IPDE extension of a bivariate continual reassessment method for jointly observed toxicity and efficacy outcomes. The model separates the intrinsic dose-level probabilities from the additional carryover contribution of an IPDE exposure. The full data from regular and IPDE patients are used to estimate the dose-response parameters and the carryover parameters r_t and r_e. Final utility is computed from the estimated joint outcome probabilities for a regular patient, because those probabilities represent the dose-level toxicity-efficacy profile after adjustment for IPDE carryover.

1 Design objective and notation

Consider ordered dose levels

$$d_1 < d_2 < \cdots < d_J,$$

with numerical dose scores x_j. For patient i treated at dose j_i, let

$$T_i = \begin{cases} 1, & \text{toxicity occurs,} \\ 0, & \text{otherwise,} \end{cases} \qquad E_i = \begin{cases} 1, & \text{efficacy occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $G_i = 0$ identify a regular patient and $G_i = 1$ identify an IPDE patient. The observed response is $(T_i, E_i) \in \{0, 1\}^2$.

For a regular patient at dose j, define the intrinsic dose-level probabilities

$$p_{Tj} = \Pr(T = 1 \mid d_j, G = 0), \qquad p_{Ej} = \Pr(E = 1 \mid d_j, G = 0).$$

These probabilities describe the current dose after separating the additional contribution associated with IPDE.

2 Marginal dose-response and IPDE carryover models

A convenient logistic specification is

$$\operatorname{logit}(p_{Tj}) = \beta_{T0} + \beta_{T1} x_j, \qquad \beta_{T1} > 0,$$

$$\operatorname{logit}(p_{Ej}) = \beta_{E0} + \beta_{E1} x_j + \beta_{E2} x_j^2.$$

A monotone efficacy model or a skeleton-based CRM model may be substituted when scientifically appropriate.

For an IPDE patient at dose j, the marginal toxicity and efficacy probabilities are

$$\Pr(T = 1 \mid d_j, G = 1) = r_t + (1 - r_t) p_{Tj},$$

$$\Pr(E = 1 \mid d_j, G = 1) = r_e + (1 - r_e) p_{Ej},$$

where $0 \leq r_t \leq 1$ and $0 \leq r_e \leq 1$.

The parameter r_t represents the residual toxicity contribution associated with the prior exposure. The parameter r_e represents the residual efficacy contribution associated with the prior exposure. When $r_t = 0$ or $r_e = 0$, the corresponding IPDE marginal is the same as the regular-patient marginal. Larger values increase the corresponding event probability for an IPDE patient treated at the same current dose.

3 Thall-Cook joint model for regular and IPDE patients

For $t, e \in \{0, 1\}$, the regular-patient joint cell probability at dose j is

$$\begin{aligned}\pi^{\mathrm{R,TC}}_{te,j} &= p_{Tj}^t(1-p_{Tj})^{1-t}p_{Ej}^e(1-p_{Ej})^{1-e} \\ &\quad + (-1)^{t+e}p_{Tj}(1-p_{Tj})p_{Ej}(1-p_{Ej})\frac{\exp(\gamma)-1}{\exp(\gamma)+1}.\end{aligned} \tag{1}$$

For an IPDE patient at dose j, the corresponding joint cell probability is

$$\begin{aligned}\pi^{\mathrm{I,TC}}_{te,j} &= \{r_t + (1-r_t)p_{Tj}\}^t\{1-r_t-(1-r_t)p_{Tj}\}^{1-t} \\ &\quad \times \{r_e + (1-r_e)p_{Ej}\}^e\{1-r_e-(1-r_e)p_{Ej}\}^{1-e} \\ &\quad + (-1)^{t+e}\{r_t + (1-r_t)p_{Tj}\}\{1-r_t-(1-r_t)p_{Tj}\} \\ &\quad \times \{r_e + (1-r_e)p_{Ej}\}\{1-r_e-(1-r_e)p_{Ej}\}\frac{\exp(\gamma)-1}{\exp(\gamma)+1}.\end{aligned} \tag{2}$$

The Thall-Cook formulation preserves the specified toxicity and efficacy marginals. The value $\gamma = 0$ gives independence; positive values increase concordant outcomes and negative values increase discordant outcomes.

4 Braun's original bCRM association model

Braun's original bCRM specifies a direct joint probability model whose cross-product odds ratio is $\psi/(1-\psi)$. For a regular patient at dose j,

$$\pi^{\mathrm{R,B}}_{te,j} = \frac{p_{Tj}^t(1-p_{Tj})^{1-t}p_{Ej}^e(1-p_{Ej})^{1-e}\psi^{te}(1-\psi)^{1-te}}{(1-\psi)(1-p_{Tj}p_{Ej}) + \psi p_{Tj}p_{Ej}}, \qquad 0 < \psi < 1. \tag{3}$$

For an IPDE patient at dose j,

$$\pi^{\mathrm{I,B}}_{te,j} = \frac{\begin{array}{c}\{r_t + (1-r_t)p_{Tj}\}^t\{1-r_t-(1-r_t)p_{Tj}\}^{1-t} \\ \times \{r_e + (1-r_e)p_{Ej}\}^e\{1-r_e-(1-r_e)p_{Ej}\}^{1-e}\psi^{te}(1-\psi)^{1-te}\end{array}}{\begin{array}{c}(1-\psi)\left[1-\{r_t + (1-r_t)p_{Tj}\}\{r_e + (1-r_e)p_{Ej}\}\right] \\ + \psi\{r_t + (1-r_t)p_{Tj}\}\{r_e + (1-r_e)p_{Ej}\}\end{array}}. \tag{4}$$

Under this model,

$$\mathrm{OR}(T, E) = \frac{\psi}{1-\psi}.$$

Thus, $\psi = 1/2$ corresponds to independence, $\psi > 1/2$ to positive association, and $\psi < 1/2$ to negative association. Unlike the Thall-Cook construction, the quantities inserted as p_{Tj} and p_{Ej} in Braun's normalized joint model are not generally equal to the resulting marginal event probabilities when $\psi \neq 1/2$. For this reason, utility should be computed from the fitted joint cells $\pi^{\mathrm{R,B}}_{te,j}$ rather than from p_{Tj} and p_{Ej} alone.

5 Combined likelihood and posterior distribution

Let $n_j^{\mathrm{R},te}$ denote the number of regular patients treated at dose j with outcome $(T, E) = (t, e)$, and let $n_j^{\mathrm{I},te}$ denote the corresponding number of IPDE patients. For either joint model $M \in \{\mathrm{TC}, \mathrm{B}\}$, the grouped likelihood is

$$L_M(\theta \mid \mathcal{D}_n) = \prod_{j=1}^{J} \prod_{t=0}^{1} \prod_{e=0}^{1} \left(\pi^{\mathrm{R},M}_{te,j}\right)^{n_j^{\mathrm{R},te}} \left(\pi^{\mathrm{I},M}_{te,j}\right)^{n_j^{\mathrm{I},te}}.$$

For the Thall-Cook model,

$$\theta_{\mathrm{TC}} = (\boldsymbol{\beta}_T, \boldsymbol{\beta}_E, r_t, r_e, \gamma),$$

and for the Braun model,

$$\theta_{\mathrm{B}} = (\boldsymbol{\beta}_T, \boldsymbol{\beta}_E, r_t, r_e, \psi).$$

With prior density $p(\theta)$,

$$p(\theta \mid \mathcal{D}_n) \propto L_M(\theta \mid \mathcal{D}_n) p(\theta).$$

Possible priors include beta distributions for r_t and r_e, a normal prior for γ, and a beta prior for ψ centered at $1/2$ when prior independence is desired.

6 Utility based on carryover-adjusted dose-level estimates

Assign clinical utilities u_{te} to the four toxicity-efficacy outcomes. After fitting the model to all regular and IPDE observations, define the posterior estimated regular-patient joint probability at dose j by

$$\widehat{\pi}^{\mathrm{R},M}_{te,j} = E\left(\pi^{\mathrm{R},M}_{te,j} \mid \mathcal{D}_n\right).$$

The dose-level posterior expected utility is

$$\widehat{U}_j = \sum_{t=0}^{1} \sum_{e=0}^{1} u_{te} \widehat{\pi}^{\mathrm{R},M}_{te,j}.$$

The final recommended dose may be selected as

$$\widehat{j}_{\mathrm{OBD}} = \arg \max_{j \in \mathcal{A}(\mathcal{D}_n)} \widehat{U}_j,$$

where $\mathcal{A}(\mathcal{D}_n)$ is a prespecified admissible set, for example one requiring acceptable posterior toxicity and efficacy probabilities.

This utility is intentionally based on the regular-patient joint probabilities rather than the IPDE joint probabilities. The IPDE observations still contribute to the likelihood and therefore affect the posterior estimates of $\boldsymbol{\beta}_T$, $\boldsymbol{\beta}_E$, r_t, r_e, and the association parameter. Estimation of r_t and r_e separates the additional carryover component from the intrinsic current-dose profile. Consequently, the fitted regular-patient probabilities are carryover-adjusted estimates of the dose-level toxicity-efficacy distribution and can be used directly for final utility calculation.

The phrase “deflated probabilities” should be used cautiously. Relative to a naive analysis that treats IPDE outcomes as ordinary dose-level outcomes, estimating positive r_t or r_e will often reduce the corresponding estimated intrinsic event probability. This direction is not guaranteed in every realized data set because all model parameters are estimated jointly. For assignment or safety evaluation of a patient already known to be an IPDE patient, the IPDE-specific probabilities in Equations (2) or (4) may still be relevant even though final OBD utility is based on the regular-patient profile.

References

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